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Master Thesis

Hardware Implementation of Gate-Current Optimization Techniques for Switching Loss and Electromagnetic Emission Reduction in SiC MOSFETs

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Study Program:	M.Sc. Electrical Engineering
Matriculation Number:	3696437
IEW Number:	IEW 4013
Starting Date:	21.07.2025
Submission Date:	20.01.2026
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Abstract

In modern power electronic applications, efficient switching of power MOSFETs is critical, requiring simultaneous minimization of switching losses and electromagnetic emissions (EME). However, conventional voltage-source gate drivers employing fixed gate resistances (R_G) offer limited flexibility in gate-current control, resulting in suboptimal switching waveforms that impose a strict trade-off between switching loss reduction and EME suppression. This limitation becomes increasingly critical for wide-bandgap devices such as silicon carbide (SiC) MOSFETs, which exhibit fast switching transitions. This thesis focuses on the practical realization of a gate-current optimization framework aimed at simultaneously minimizing switching energy and EME in real hardware implementations. Initially, the theoretical foundations of switching loss and EME in power MOSFETs are established. Subsequently, the state-space simulation framework employed in this work is described and adapted for a Bosch SiC MOSFET by incorporating experimentally extracted device characteristics and parasitic parameters. Following that, the definitions and formulas of the switching loss and EME-related metrics (Heisenberg-Gabor Inequality (HGI)) used in this work are introduced. Baseline simulations using fixed gate resistances are performed to quantify the inherent limitations of conventional gate-driving methods with respect to switching loss and EME reduction. Thereafter, a gate-current optimization framework based on a genetic algorithm is integrated into the simulation environment. The optimization employs a multi-objective cost function that quantitatively balances switching loss and EME while constraining the drain-source voltage (V_{DS}) overshoot within specified limits. The key contribution of this thesis lies in the experimental realization of these optimized gate-current profiles using a programmable current-source gate driver (EG120). Detailed experimental validation is conducted on a custom-built double pulse testing board, during which the hardware was operated at peak drain currents of up to 275 A and peak drain–source voltages of up to 1000 V. The experimental results of the optimized gate-current profile approach demonstrate significant improvements over conventional fixed gate-resistance methods. The loss-optimized EG120 configuration shows a reduction in the low-side device switching loss by 14.4% (vs. 5.1 Ω R_G case) and 42.6% (vs. 7.5 Ω R_G case), while simultaneously lowering the high-side device HGI by 21.7% and the V_{DS} overshoot by 47.4%. Furthermore, the EME-optimized EG120 configuration achieves a 58.6% reduction in the high-side HGI and a 59.7% reduction in V_{DS} overshoot compared to the fixed 5.1 Ω R_G baseline. Observed discrepancies between simulated and measured results are systematically analyzed, highlighting the limitations of the current source driver employed. Overall, the results of this work demonstrate the strong potential of active gate drivers, capable of delivering tailored gate-current profiles. By demonstrating the practical feasibility of optimized gate-current shaping in real hardware, this thesis bridges the gap between theoretical optimization and experimental realization, enabling improved efficiency and EME behavior in power electronic converters.

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6 Summary and Conclusion

6.1 Summary

This thesis successfully demonstrated the experimental validation of a gate-current optimization framework for SiC MOSFETs, enabling a significantly reduced trade-off between switching losses (E_{sw}) and electromagnetic emissions (EME) compared to conventional fixed gate-resistance driving. Each of the four major research questions covering the definition of optimal switching behavior, the application of genetic algorithms (GA) for gate-current optimization, the experimental hardware implementation of the optimized gate-current profiles, and their associated limitations were comprehensively investigated, and successful results were obtained.

A physics-based state-space simulation framework was introduced and calibrated using experimentally extracted device parameters, including nonlinear device capacitances (C_{GS}, C_{GD}, C_{DS}), drain-current (i_D), and loop inductance (L_σ), which was subsequently employed to perform double-pulse test (DPT) simulations of the chosen Bosch SiC MOSFET. The fixed gate-resistance simulation results were then visually compared with experimental DPT measurements to verify that the simulation framework sufficiently captures the real-world switching behavior and can therefore be used for the subsequent optimization study.

A GA–based optimization framework was then used to derive optimal gate-current profiles. Two distinct optimization methodologies were investigated: a continuous gate-resistance optimization approach used for theoretical benchmarking, and a discrete gate-current optimization approach that explicitly incorporates some practical constraints of the EG120 programmable gate driver, including limited gate current amplitude and discrete time intervals. This optimization employed a multi-objective cost function that quantitatively balances switching loss and EME (quantified using the Heisenberg–Gabor Inequality (HGI)), while ensuring drain-source voltage overshoot constraints.

Simulations employing the optimized gate-current profiles demonstrated clear performance improvements compared to conventional fixed gate-resistance operation. In this thesis, E_{sw} is evaluated for the LS MOSFET during turn-on, while EME is assessed based on the v_{DS} waveforms of both the LS and HS devices. For the EME-optimized case ([1,9]), the HS v_{DS} exhibited a waveform shape closer to a Gaussian function with negligible overshoot and ringing, resulting in an $\mathrm{HGI}_{\mathrm{HS}}$ value of 0.53, corresponding to a 91.5% reduction compared to the fixed $R_G = 5.1\,\Omega$ case, while also achieving a 12% reduction in switching loss. In contrast, the switching-loss-optimized case ([9,1]) achieved the lowest switching loss among all evaluated scenarios (1396.7 μJ), corresponding to a 50.2% reduction compared to the fixed $R_G = 5.1\,\Omega$ case, but at the expense of degraded EME performance.

The theoretically derived gate-current profiles were applied using the EG120 current-source gate driver on a custom-designed double-pulse test (DPT) hardware. During the experiment, the hardware experienced peak drain currents of up to 275 A and peak drain–source voltages

of up to 1000 V, with the option to introduce external commutation inductance of up to 80 nH via configurable jumper connections. At the operating point of L_σ = 20 nH, V_{DC} = 850 V, and I_{Load} = 175 A, the EG120 configuration optimized for switching-loss reduction ([9,1]) achieved a minimal E_{sw} of 2209 μJ, corresponding to a relative reduction of 14.4% compared to the fixed gate-resistance case with the lowest switching loss (R_G = 5.1 Ω). Moreover, it also exhibited reduced EME metrics, with a 21.7% reduction in $\mathrm{HGI}_{\mathrm{HS}}$ and a 47.4% reduction in the V_{DS} voltage overshoot. When benchmarked against the 7.5 Ω fixed R_G case, the same EG120 configuration yielded a switching-loss reduction of 42.6% while maintaining a comparable EME performance, with only marginal degradation.

Moreover, the EME-optimized EG120 configuration ([1,9]) achieved the best overall EME performance among all investigated experiment cases, with $\mathrm{HGI}_{\mathrm{LS}}$ = 10.16 and $\mathrm{HGI}_{\mathrm{HS}}$ = 10.58. This corresponds to a 58.6% reduction in $\mathrm{HGI}_{\mathrm{HS}}$ and a 59.7% reduction in the V_{DS} overshoot compared to the 5.1 Ω fixed R_G configuration. These experimental results confirmed that the implemented EG120 gate-current profiles successfully reproduced the intended optimization objectives, despite practical hardware constraints.

Observed discrepancies between simulated and measured results were systematically analyzed and were attributed to the finite current rise times, internal delays, and limited maximum current-providing capability of the EG120 driver, which prevented the exact reproduction of the ideal simulated gate-current profiles. Despite these hardware limitations, the results demonstrated that optimized gate-current driving significantly outperforms conventional fixed R_G operation, effectively reducing the traditional efficiency–EME trade-off in modern power electronic converters.

6.2 Conclusion

The experimental validation in this work was conducted with L_σ of approximately 20 nH. In industrial power electronic systems employing complex converter topologies, such as multilevel inverters, more intricate busbar structures inevitably result in significantly higher L_σ [37]. In such high-inductance regimes, voltage overshoot becomes the primary limitation on switching speed, leading to increased switching losses. Under these conditions, the optimized gate-current profiling method employed in this thesis enables selective shaping of the switching transient, rather than uniformly slowing the entire commutation process. This indicates that the performance advantage of the gate-driving method discussed in this thesis becomes even more pronounced in high-inductance power electronic systems.

While theoretically optimal gate-current profiles can substantially reduce switching losses and EME, their practical realization is often constrained by the limitations of commercially available current-source gate drivers, such as the EG120. Future research may therefore focus on bridging the gap between ideal continuous-resistance profiles or current-source gate-driving concepts and practical voltage-source implementations, thereby avoiding the need for dedicated current-source drivers. Discrete analog closed-loop approaches, such as using an auxiliary MOSFET in parallel with the gate resistor [38] or external analog control circuits based on differentiators and high-speed comparators with $\mathrm{d}v_{DS}/\mathrm{d}t$ feedback [8], can enable quasi-continuous gate-current or gate-resistance profiles that approximate towards the theoretical optimum. However, the increased circuit complexity of such approaches may limit their suitability for cost and reliability-driven industrial applications.

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